

Credit Card Fraud Detection

Jakub Karłowski

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Abstract

This report evaluates and optimizes classical ensemble methods alongside deep learning architectures to tackle extreme class imbalance in credit card fraud detection. Using SMOTE for supervised models and an unsupervised anomaly detection approach, we build a reliable classification system evaluated thoroughly through Precision-Recall metrics.

1 Introduction & Objective

Credit card fraud detection is a critical challenge for modern financial institutions, requiring automated systems capable of identifying unauthorized transactions in real time. This project evaluates and compares various machine learning methodologies using the standard Credit Card Fraud Detection benchmark dataset, originally collected and curated by the Machine Learning Group at Université Libre de Bruxelles (ULB). The dataset contains real, anonymized transactions made by European cardholders over a two-day period in September 2013.

The primary challenge of this dataset is its extreme class imbalance: out of 284,807 transactions, only 492 are classified as fraud, representing a mere 0.172% of the data. In such scenarios, standard metrics like accuracy become completely misleading. For example, a naive baseline model that constantly predicts a transaction is legal will achieve an accuracy of 99.828%, yet it fails entirely to detect any actual security threats.

To find the best model for actual banking systems, we handle the class imbalance using SMOTE and unsupervised learning, evaluating the final results with Precision-Recall curves to choose the most practical model for real-world deployment.

2 Data Preprocessing & Methodology

2.1 Feature Scaling & Data Splitting

Since the dataset contains anonymized components derived from Principal Component Analysis (PCA), denoted as V1 to V28, only the Time and Amount features required standardization. A StandardScaler was applied to transform these features, ensuring that large numerical magnitudes do not dominate distance-based or gradient calculations. The dataset was then split into an 80/20 training and testing ratio using stratified sampling (`stratify=y`) to maintain identical fraud proportions across both subsets.

2.2 Handling Class Imbalance

To address the severe class imbalance, two distinct strategies were implemented:

- **Supervised Models:** The training data was balanced using SMOTE (Synthetic Minority Over-sampling Technique). SMOTE synthesizes new examples of the minority class (fraud)

along the line segments joining k-nearest neighbors, bringing the class ratio to a perfect 50/50 split for training.

- Unsupervised Model (Autoencoder): To provide a robust baseline for anomaly detection, the Autoencoder was strictly trained only on genuine transactions (X_train_legal), entirely bypassing the SMOTE algorithm.

3 Model Architectures

The project comprehensively evaluates eight different models, classified into three main categories:

- Baseline Model:
 - Single Decision Tree: A standalone estimator restricted to max_depth=5 to act as a benchmark performance control.
- Ensemble Methods:
 - Random Forest: A bagging-based ensemble utilizing 100 deep decision trees, optimized via multi-core CPU parallelization (n_jobs=-1).
 - XGBoost: An advanced, highly optimized gradient boosting framework designed for speed and performance execution.
 - AdaBoost: A sequential boosting ensemble using shallow base estimators (max_depth=2) with a tuned learning rate of 0.1.
- Neural Networks & Deep Learning:
 - Basic MLP (Sklearn): A shallow Multi-Layer Perceptron containing two hidden layers with 32 and 16 neurons, respectively.
 - Deep MLP (Keras): A deep neural network optimized with BatchNormalization layers to accelerate convergence and Dropout(0.3) layers to mitigate overfitting.
 - 1D CNN (Keras): A 1D Convolutional Neural Network designed to extract structural, localized correlations among PCA-transformed variables.
 - Autoencoder (Keras): An unsupervised neural network utilizing an architectural bottleneck (compressing dimensions down to 8 neurons) to model standard behavior and catch fraud through reconstruction errors (MSE).

4 Empirical Results & Discussion

4.1 Precision-Recall Analysis

We use the Precision-Recall (PR) Curve instead of the traditional ROC curve. Because the dataset has so few fraud cases, a ROC curve would look misleadingly perfect just because the number of true negatives is huge. The PR curve focuses directly on the fraud class.

The comparative ranking based on Average Precision (AP) taken directly from our empirical results is structured as follows:

1. Random Forest: AP = 0.8479
2. XGBoost: AP = 0.8374
3. Deep MLP: AP = 0.8319
4. Basic MLP: AP = 0.8069

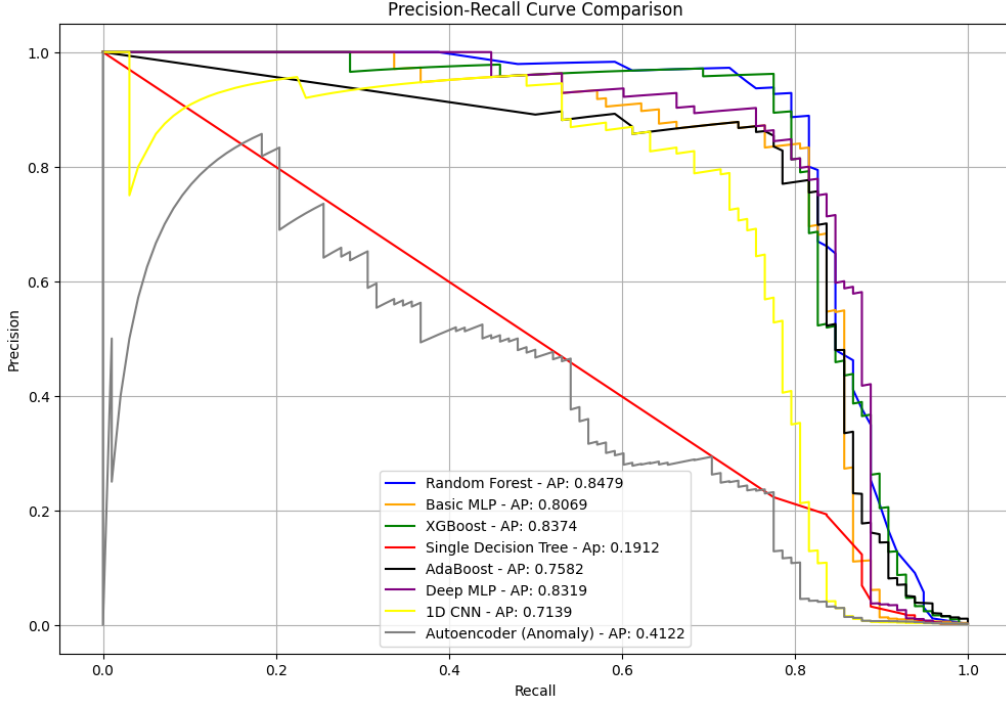


Figure 1: Precision-Recall Curve Comparison (Sorted by AP)

5. AdaBoost: AP = 0.7582
6. 1D CNN: AP = 0.7139
7. Autoencoder (Anomaly): AP = 0.4122
8. Single Decision Tree: AP = 0.1912

4.2 Computational Performance

The execution times recorded during the training phase are detailed in Table 1.

Table 1: Computational Training Durations and Metric Scores

Model Architecture	Training Duration (Seconds)	Average Precision (AP)
Random Forest	430.50 s	0.8479
Basic MLP (Sklearn)	55.48 s	0.8069
XGBoost	10.98 s	0.8374
Single Decision Tree	19.24 s	0.1912
AdaBoost	817.09 s	0.7582
Deep MLP (Keras)	49.80 s	0.8319
1D CNN (Keras)	270.69 s	0.7139
Autoencoder	10.26 s	0.4122

While ensemble frameworks like Random Forest make exceptional use of multi-core CPU parallelization, they still take over 7 minutes to train on the oversampled dataset. Sequential boosting via AdaBoost proved to be the most expensive, taking 817.09 seconds.

In contrast, XGBoost demonstrated incredible computational efficiency, finishing the training phase in just 10.98 seconds while achieving an impressive AP score of 0.8374. Deep MLP

implemented in Keras also showed a competitive profile, training in under 50 seconds due to large mini-batch execution (batch_size=1024).

4.3 Confusion Matrix Evaluation

To understand the financial implications of model selection, the confusion matrices of the top three performing models (Random Forest, XGBoost, and Deep MLP) were analyzed based on 98 actual fraud instances in the test partition.

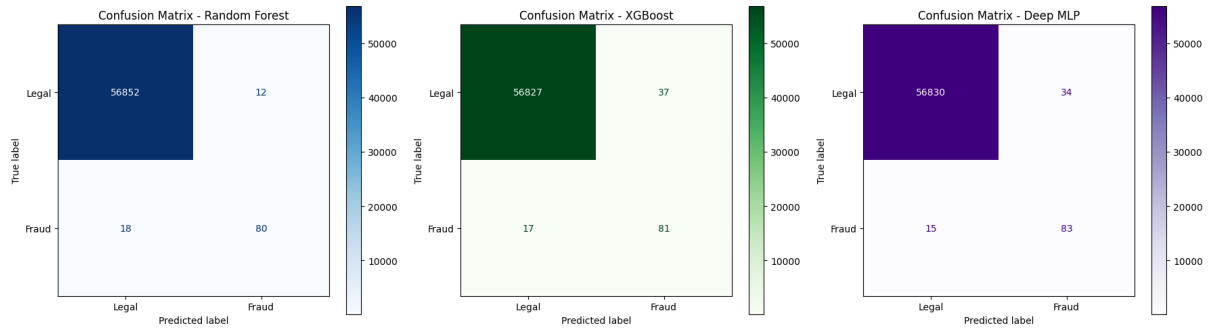


Figure 2: Comparative Confusion Matrices (RF, XGBoost, Deep MLP)

- False Negatives (Missed Frauds): Deep MLP excelled by minimizing financial leaks, allowing only 15 undetected fraudulent charges (catching 83 out of 98), compared to 17 missed by XGBoost and 18 missed by Random Forest.
- False Positives (False Alarms): Random Forest proved to be the most customer-friendly model, causing only 12 false alarms out of over 56,000 legal transactions, significantly reducing customer friction compared to 37 false alarms from XGBoost and 34 from Deep MLP.

From a business perspective, XGBoost offers arguably the best trade-off in the entire project. It missed only 17 frauds (compared to 18 by Random Forest) and triggered 37 false alarms, meaning it delivers top-tier security performance at a fraction of the computational cost.

5 Feature Importance

To understand how the models make decisions, we analyzed feature importance. We used Gini importance for the tree-based models and Permutation Feature Importance for the Deep MLP.

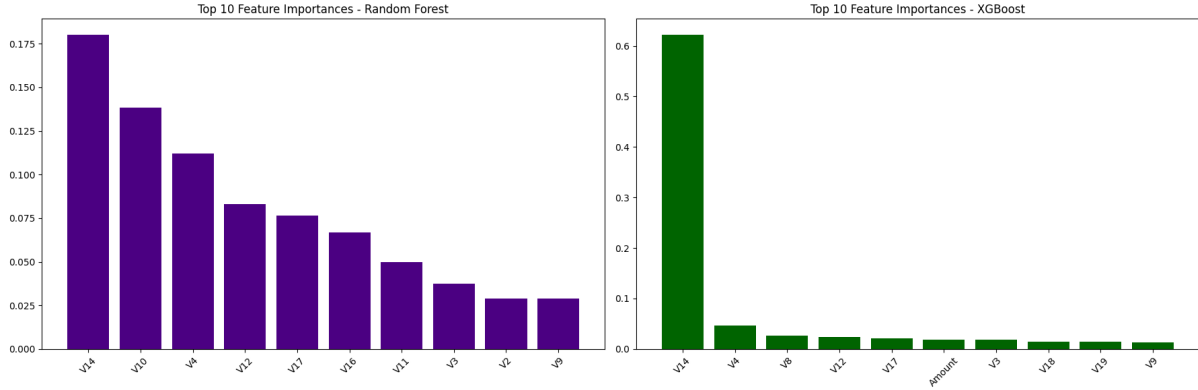


Figure 3: Top 10 features for Random Forest and XGBoost based on tree-based importance metrics.

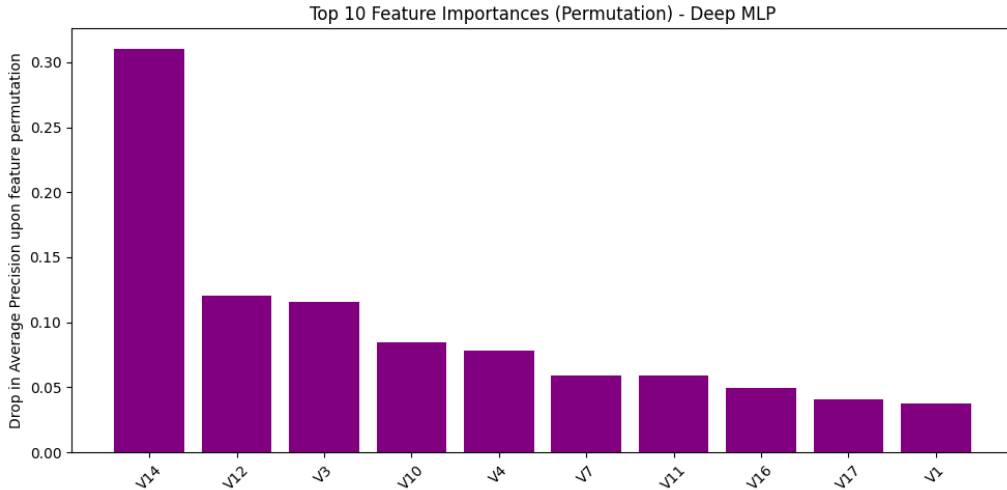


Figure 4: Top 10 features for the Deep MLP model based on Permutation Importance on the test set.

Despite using completely different approaches—tree splitting versus neural network gradients—all top models relied on the same key features. V14, V12 and V10 consistently appeared at the top of the charts, showing they have the strongest predictive power for fraud detection.

6 Conclusion

This project successfully demonstrated the deployment of advanced machine learning for credit card fraud detection.

- Supervised Winner: While Random Forest achieved the absolute highest AP score (0.8479) and the lowest false alarm rate, XGBoost emerges as the most practical model for real-world production. Training in just 10.98 seconds with an AP of 0.8374, while missing only 17

frauds, XGBoost provides an unmatched balance of speed and predictive power. It is highly recommended for banking systems that require frequent model retraining on fresh data. If absolute fraud containment is the only priority regardless of training overhead, the Deep MLP remains a strong alternative due to its 15 missed frauds.

- The Role of the Autoencoder: Although the unsupervised Autoencoder achieved a lower overall AP score (0.4122), its implementation remains vital. Because it models normality rather than historical fraud signatures, it serves as the institution’s primary defense line against zero-day exploits—novel fraud methodologies that have never been captured in historical records.